FAYEZA AFRAH HISSAN

B.Sc. in Computer Science & Engineering, IIUC

Email: f.a.hissan1311@gmail.com
Phone: +8801976504337
Location: Chittagong, Bangladesh

OBJECTIVE

Final-year Computer Science and Engineering student with a decent CGPA, 13 academic projects, two published IEEE papers, and tutoring experience. Final thesis on graph neural networks completed. Strong foundation across CSE courses and a proven ability to explain concepts clearly to students and peers, working towards a career in academia and research.

EXPERIENCE

Teaching Assistant, IIUC
Currently working as a Teaching Assistant at International Islamic University Chittagong.

Private Tutor (SSC level) | *about 3–4 years*
Strengthened fundamentals and prepared students for board exams.

Peer Mentor | *undergraduate years*
Group studies, explaining course topics to classmates, and reviewing and debugging team projects and reports.

ACHIEVEMENTS, PUBLICATIONS & CERTIFICATIONS

Achievements

- Author and Presenter, QPAIN 2026 (IEEE International Conference on Quantum Photonics, AI & Networking)
- Champion, Intra-University Math Olympiad, IIUC Tech Fest 2025
- 1st Runner-up, Intra-University Math Olympiad, IIUC Tech Fest 2023

Publications

1. **F. A. Hissan**, A. M. Jahan and S. Karim, "Enhancing LSB Image Steganography Through Password Protection and Constraint Satisfaction-Based Message Filtering," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1–6, doi: 10.1109/QPAIN69676.2026.11546392.

2. A. M. Jahan, I. J. Worthy and **F. A. Hissan**, "Multi-Model Insider Threat Detection on Temporally Aggregated and Imbalanced Data: A Comparative Study Using SMOTE and Feature Ablation," *2026 IEEE QPAIN*, Chittagong, Bangladesh, 2026, pp. 1–6, doi: 10.1109/QPAIN69676.2026.11546056.

Certifications

- Student Delegate, ICISET 2024 (International Conference on Innovation in Science, Engineering and Technology), IIUC
- Microsoft Certified: Ask Questions and Analyze Content with Microsoft 365 Copilot

Research profiles

ResearchGate: researchgate.net/profile/Fayeza-Afrah-Hissan
Google Scholar: scholar.google.com/citations?user=azK6364AAAAJ
GitHub: github.com/FAH-works

EDUCATION

B.Sc. in Computer Science and Engineering 2022 – 2026 (Expected)
International Islamic University Chittagong

- **Current CGPA: 4.00** | Final semester completed | Thesis completed
- Results pending: Thesis, Field Work, General Viva

Higher Secondary Certificate (HSC) 2021 | GPA 5.00
Chattogram Cantonment Public College

Secondary School Certificate (SSC) 2019 | GPA 5.00
Bangladesh Mahila Samiti Girls' High School and College

SKILLS

Programming Languages: C, C++, Python, PHP

Web & Databases: HTML, CSS, Django, MERN, Oracle Database

Tools & Platforms: Google Colab, Kali Linux, Arduino, Codeblocks, XAMPP, LaTeX, MS Word, MS PowerPoint

Relevant Areas: Algorithms, Operating Systems, Numerical Methods, Software Engineering, Machine Learning, Neural Networks, Image Processing, Computer Security

Competitive Programming: Codeforces (155+ problems solved), UVa Online Judge (56+ problems solved)

Languages: Bangla (Native), English (Fluent)

Others: Strong written & verbal communication, fast and accurate typing, effective AI-prompting, team collaboration, reviewing & debugging of team projects, punctuality with deadlines

PROJECTS

Make Me Pretty *Web Development | Team project (Field Work)*
MERN-stack e-commerce website for skincare, body care, hair care and makeup products, with customer, admin and delivery dashboards.

UrbaSafe *Image Processing*
Saliency-guided framework analyzing urban imagery for safety perception and renewal (Google Colab).

Hide & Seek *Web / Security*
LSB steganography messaging platform (HTML, CSS, Python, Django) with password protection and constraint satisfaction-based filtering; first-author IEEE paper.

Insider Threat Detection on Temporally Aggregated Data *Machine Learning*
Random Forest, XGBoost and SVM with SMOTE and feature ablation on imbalanced data; co-authored IEEE paper.

Smart Home Network *Networks / IoT*
OMNeT++ smart-home network (TV, AC, fridge, printer, washing machine) with ML-based QoS monitoring and IoT traffic classification.

Fuzzy SVM vs. Standard SVM *Neural Network & Fuzzy System*
Compared standard SVM against Fuzzy SVM across five financial classification datasets (Google Colab).

Port Knocking Based Firewall *Computer Security*
Port-knocking firewall on Kali Linux (VirtualBox) for a Computer Security course project.

Zombie Apocalypse OS *Operating Systems*
Simulated C++ survival kernel with Banker's Algorithm, Variable Best-Fit allocation and Preemptive Priority Scheduling.

Future Focus *Algorithms*
C++ major-mapping assistant using the Knapsack algorithm, based on skills, interests and budget.

The Automated Traffic Systemizer *Microprocessor / EEE*
Arduino traffic control model with timed lights, automatic barrier, buzzer and streetlights.

Nutri-Track Academia *Web Development*
Student meal-management system (PHP, HTML, CSS): ordering, billing and BMI/BMR-based personalized menus.

Immunify *Web Development*
COVID-19 vaccine registration system (PHP, HTML, CSS) hosted on XAMPP for appointment booking.

Flavors & Cravings *Database Management*
Oracle database for a multi-store food distributor: pricing, orders and supply.